

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

SKILL-1:

Aim: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parentchild relationships, inspect process tree, practice system call tracing.

```
venkatsaikrishna5@VSK-HP:~$ cd ~
venkatsaikrishna5@VSK-HP:~$ mkdir LinuxShellProject
venkatsaikrishna5@VSK-HP:~$ cd LinuxShellProject
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ mkdir src include
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ touch Makefile README.md
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ touch src/main.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ ls -R
.:
Makefile README.md build docs include src tests
./build:
./docs:
./include:
./src:
main.c
./tests:
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ nano src/main.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ nano Makefile
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ make
gcc -Wall src/main.c -o shell
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ ./shell
Linux Shell Project Started!
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ ps
  PID TTY          TIME CMD
   339 pts/0    00:00:00 bash
   633 pts/0    00:00:00 ps
```

```
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ pstree
systemd
├──agetty
├──chronyd-starter─chronyd─chronyd
├──cron
├──dbus-daemon
├──init-systemd(Ub)─SessionLeader─Relay(339)─bash─+
++
├──networkd-dispat
├──rsyslogd─3*[{rsyslogd}]
├──systemd─(sd-pam)
├──systemd-journal
├──systemd-logind
├──systemd-resolve
├──systemd-udevd
├──unattended-upgr─{unattended-upgr}
├──wsl-pro-service─7*[{wsl-pro-service}]
├──init─{init}
├──login─bash
└──init-systemd(Ub)
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ nano src/fork_demo.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ gcc src/fork_demo.c -o fork_demo
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ ./fork_demo
Hello
Hello
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ nano src/exec_demo.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ gcc src/exec_demo.c -o exec_demo
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ ./exec_demo
Makefile build exec_demo include src
README.md docs fork_demo shell tests
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ nano src/parent_child.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ gcc src/parent_child.c -o parent_child
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ ./parent_child
I am Parent Process
I am Child Process
```


Code:

```
#include <stdio.h>

int main()
{
    printf("Linux Shell Project Started!\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if(pid == 0)
        printf("I am Child Process\n");
    else
        printf("I am Parent Process\n");

    return 0;
}
```

```
#include <unistd.h>

int main()
{
    execl("/bin/ls", "ls", NULL);
    return 0;
}
```

Observation :

- A Linux virtual machine was installed and configured successfully.
- GCC was configured and used to compile C programs.
- A Git repository and basic project structure were created.
- The shell architecture and basic Makefile structure were understood.
- Process abstraction was studied using `fork()`.
- The `exec()` family was used to replace a process with a new program.
- Parent-child relationships were observed using process IDs and the process tree.
- System calls generated by processes were inspected using tracing tools.

Result :

The Linux development environment was successfully prepared with GCC, Git, project structure, and Makefile. Process management concepts were demonstrated using `fork()` and `exec()`, parent-child relationships and process trees were analyzed, and system call tracing was successfully performed to understand how processes interact with the Linux kernel.